

Yaz: an editable, auditable tiny knowledge model with routing-confidence abstention

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Abstract

We present **Yaz**, a sub-1M-parameter, byte-level language model designed so that individual facts can be created, read, updated, and deleted (CRUD) one at a time, with **provable per-edit locality**, and so that the model **abstains** when it is not confident which fact a prompt refers to. Each fact is stored in its own addressable decoder column (an *atom*); prompts are routed to atoms by meaning using a frozen sentence embedding, so paraphrases reach the same fact. We report what is verifiable on a small synthetic benchmark (50 country→capital facts): exact in-distribution edit reliability, zero-collateral deletion, structural locality across 40 sequential edits (bits-per-character change +0.000%), paraphrase-robust routing (held-out reach 0.696 vs 0.216 for surface routing), and a near-oracle abstention signal (risk-coverage AURC 0.004 against an oracle bound of 0.003). We are equally explicit about what Yaz does **not** do: it is a *first-byte* editor (multi-byte generation is not faithful; full-word transfer ≈ 0.05), routing degrades on oblique name-free clues, and the scope is tiny and synthetic. Yaz is a clean, reproducible prototype in the established model-editing lineage (ROME/MEMIT, GRACE, SERAC, PENME), **not** a state-of-the-art result and not a defensible new capability. Its contribution is engineering clarity.

1 Motivation

Large models store facts diffusely across billions of weights. Two consequences follow: editing one fact risks perturbing others, and a confidently wrong answer is indistinguishable from a correct one. Yaz is a deliberately small experiment in the opposite design: put each fact in its own slot, make edits surgical and reversible, and let the model decline when it is unsure which fact is being asked about. The goal is *auditability*, not capability.

2 Method

Fact-atoms. Yaz is a byte-level transformer whose output head is augmented with an additive *FactAtomLayer*: fact i owns decoder column i . When a prompt is routed to fact i , that column’s contribution biases the first generated byte toward the stored answer.

Routing by meaning. A frozen sentence embedder (`sentence-transformers/all-MiniLM-L6-v2`, 384-d, L2-normalized) maps a prompt to a vector. Each fact’s *centroid* is the mean embedding of

its training-template phrasings; a prompt routes to the nearest centroid by cosine similarity. The embedder is never fine-tuned on the facts, and centroids are built only from training templates — held-out paraphrases never enter fitting, so routing generalization is measured leak-free.

CRUD verbs. UPDATE swaps one decoder column; DELETE zeroes an atom; CREATE allocates a fresh atom; READ is just routing. None require gradient steps.

Abstention. The routing *confidence margin* (top-1 minus top-2 centroid cosine) is used as a selective-prediction signal: below a threshold the model refuses rather than emitting a guess.

3 Results (verified, reproducible)

Benchmark: 50 country→capital facts, single seed (2026), CPU. All numbers reproduce exactly with the public sentence-transformers embedder (no internal dependencies).

Capability	Measured result
UPDATE (edit, no retraining)	in-distribution reliability 1.000; edits land 8/8 (first byte)
DELETE	fact removed, 0 collateral on other facts
CREATE	passes 4/4-condition battery (monosemantic / local / readable / deletable)
Per-edit locality	0/10 collateral; bpc change +0.000% across 40 sequential edits
Sequential robustness	retention flat (1.000) over 40 edits
Paraphrase-robust routing	held-out reach 0.696 vs 0.216 surface (lexical) routing
Abstention (all, $n=298$)	risk-coverage AURC 0.004 (oracle 0.003; grace 0.052; random 0.087)
Abstention operating point	coverage 0.966 at $\leq 5\%$ answered-risk (vs 0.581 grace-distance)
Abstention (hard-only, $n=48$)	AURC 0.194 (oracle 0.159; grace 0.436; random 0.591)

4 Limitations and honest scope

These are not footnotes; they bound every claim above.

- **First-byte editor.** Edits reliably set the answer’s *first byte*. Multi-byte generation is not faithful — full-word transfer is ≈ 0.05 .
- **A retracted claim.** An earlier “edit-generalization” headline of 0.675 was **retracted**: a random-column-swap control produces a null of ≈ 0.688 , so that first-byte number sits *at chance*. What survives is routing *reach*, not an edit-magnitude effect.
- **Fragile routing.** Reach is ≈ 0.85 on famous entities but falls to ≈ 0.50 on oblique, name-free clues; answered coverage drops accordingly.

- **Structural, not unconditional, locality.** The guarantee holds while no two facts route to the same atom; collisions can appear at larger fact counts.
- **Tiny, synthetic scope.** 50 facts, one seed, CPU. Not validated at scale or on open-vocabulary knowledge.
- **Not a moat.** The mechanisms each exist in the literature (§6). Yaz combines them cleanly and reproducibly; that is engineering, not a defensible capability.

5 Reproducibility

```

pip install -r requirements.txt
python scripts/scaling/s3_route_abstain.py      # abstention risk-coverage (AURC)
python scripts/scaling/s4_create_primitive.py    # the CREATE 4-condition battery
pytest -q                                       # in-dist reliability 1.000 + delete locality

```

All runs are deterministic (seed 2026), CPU-only. The routing/abstention numbers were regenerated end-to-end with the public `all-MiniLM-L6-v2` embedder, with no local paths or private packages, and matched the reported values exactly.

6 Related work

Yaz sits in the knowledge-/model-editing lineage: locate-and-edit methods (ROME, MEMIT), codebook/adaptor editors (GRACE), retrieval-routed editors (SERAC), and embedding-keyed editing (PENME, arXiv:2408.10411), with EasyEdit consolidating the family. The abstention component draws on selective prediction and risk-coverage analysis. Yaz’s only novel framing is the integration: routing-confidence used directly as a refusal signal inside a CRUD editor with structural locality.

7 Conclusion

Yaz is an honest, small, reproducible demonstration that fact-level CRUD editing, structural locality, and confidence-based abstention can coexist in a sub-1M-parameter byte-level model. It is a step, not a breakthrough. We release it as a dated, open, reproducible artifact.

License: MIT. Code: <https://github.com/TilelliLab/yaz> Weights: <https://huggingface.co/TilelliLab/yaz>